Urvish Shah

Buffalo, NY, USA | Open to Relocation | Available Immediately

+1 (716) 553-8396 shahurvish2001@gmail.com linkedin.com/in/urvish-r-shah Portfolio GitHub Google Scholar

Professional Summary

Robotics Systems Engineer with hands-on experience building, integrating, and debugging real robotic systems across **ROS2, embedded hardware, motion control, PCB bring-up, sensor integration, and sim-to-real validation**. Startup-ready full-stack builder with proven execution in **multi-robot coordination, LoRa-based localization, control systems, field testing, and hardware-software failure recovery**. Strong fit for early-stage robotics teams needing engineers who can move across software, hardware, testing, and deployment with minimal ramp-up.

Education

University at Buffalo, State University of New York — **Aug 2024 – Dec 2025**
Master of Science in Robotics, GPA: 3.63 / 4.0 — *Buffalo, NY*

Government Engineering College, Gandhinagar — **Jun 2019 – May 2023**
Bachelor of Engineering in Instrumentation and Control Engineering — *Gandhinagar, India*

Experience

ADAMS Lab, University at Buffalo — **Aug 2024 – Jan 2026**
Robotics Systems Engineer | ***Thesis:*** *Decentralized Multi-Robot Collaborative Transport and Source Localization* — *Buffalo, NY*

- Enabled **4x larger and 2x heavier object transport** than single-robot capability by architecting and deploying a decentralized **ROS2 multi-robot coordination system** using **4 TurtleBot3 robots**.
- Achieved stable cooperative transport on ramps up to **25°** by implementing distributed closed-loop control, coordinated force distribution, and fault-aware robot role allocation.
- Built real-time **Python/C++ ROS2 nodes** for sensing, control, telemetry, monitoring, and inter-robot coordination under communication dropouts and latency constraints.
- Engineered **adaptive force distribution and role allocation algorithms**, improving robustness during partial robot faults, terrain changes, unstable object motion, and varying friction/load conditions.
- Drove **Gazebo-to-real validation and system commissioning** by resolving communication loss, actuator inconsistency, control instability, and hardware-software integration failures prior to field testing.
- Integrated **Bayesian source localization using Gaussian Processes with LoRa RSSI sensing**, enabling decentralized signal estimation in uncertain and noisy environments.

Indian Institute of Technology Gandhinagar — **Jan 2023 – Apr 2024**
Project Assistant, IITGN Robotics Lab — *Gandhinagar, India*

- Led development of a **pellet-based additive manufacturing platform** for recycled plastics, enabling direct printing from granules instead of filament and supporting sustainable material reuse.
- Designed and built a **custom pellet-fed extrusion system with integrated shredding**, continuous feed operation, thermal regulation, and material-flow control.
- Developed **embedded control electronics and firmware** for extrusion, heater regulation, material flow, and process monitoring, improving repeatability and print consistency across extended runs.
- Conducted **system benchmarking, validation, and root-cause analysis**, identifying failure modes and improving uptime, robustness, and reliability under varying material conditions.
- Converted experimental system performance into peer-reviewed research outputs in recycled polymer additive manufacturing and sensor-based process monitoring.

OoB Services — **Jun 2022 – Aug 2022**
Hardware Design Engineer Intern — *Ahmedabad, India*

- Designed and validated **multi-layer PCBs in Altium Designer** for integrated sensing, controller, and power electronics systems.
- Integrated **ARM-based microcontrollers, sensors, wireless modules, and communication interfaces** including I2C, SPI, and UART into functional embedded systems.
- Performed **board bring-up, signal validation, and debugging using oscilloscopes and lab instrumentation**, resolving hardware issues and ensuring stable system operation.
- Reduced hardware iteration risk by identifying and correcting design, interface, and signal-integrity issues during early validation stages.

GTU ABU Robocon Team — **Aug 2020 – May 2023**
Robotics Hardware & Control Engineer — *Ahmedabad, India*

- Represented India at **ABU Robocon 2021**, achieving **9th place internationally** through rapid hardware iteration, controls integration, and on-site system debugging.
- Designed and implemented **embedded motor and actuator control systems using Arduino and STM32** for competition-grade robotic platforms with closed-loop operation.
- Developed **custom motor-driver and actuator-control PCBs** integrating encoders, IMUs, motors, and distance sensors for feedback-driven control and system responsiveness.
- Led **full-system integration and competition deployment** under time-critical constraints, improving reliability during live runs.
- Mentored junior engineers on hardware bring-up, debugging workflows, and structured development practices; co-invented a **patented robotic mechanism (IN202321008858)**.

Selected Projects

Scarecrow 2.0: Agricultural Monitoring System

- Designed a multi-sensor embedded system integrating **PIR, GSM, rain, moisture, and environmental sensors** for autonomous outdoor agricultural monitoring.
- Developed embedded detection and alerting logic with low-power strategies, renewable energy integration, and long-duration field operation.
- Validated system behavior through field testing and iterative tuning under variable outdoor operating conditions.

Technical Skills

Robotics & Systems: ROS, ROS2, distributed systems, system integration, motion control, real-world testing, sim-to-real validation
Programming: C++, Python, Embedded C
Embedded & Hardware: PCB Design (Altium), ARM, AVR, ESP32, sensors, actuators, microcontrollers, board bring-up, hardware debugging
Control & Algorithms: Closed-loop control, Bayesian inference, Gaussian Processes, decentralized coordination, role allocation
Tools & Simulation: Gazebo, RViz, Linux, Git, Docker, SolidWorks, 3D printing
Communication Protocols: I2C, SPI, UART, LoRa

Publications

Multi-Robot Box Transport Over Different Surfaces With Decentralized Role-Based Proportional Control. *ASME IDETC/CIE*, 2026. **Accepted**.

A Digital Twin Framework for Autonomous UAV Tracking with Gazebo-Based Simulation and Sim-to-Real Validation. *AIAA SciTech Forum*, 2026.

Sustainable Recycling of ABS: Thermophysical Characterisation in Filament vs. Granules-Based 3D Printing. *Rapid Prototyping Journal*, 2025.

Multi-Sensor Deep Learning Framework for Detection of Nozzle Clogging in Pellet-Based 3D Printing. *Progress in Additive Manufacturing*, 2025.

Achievements

Represented India at ABU Robocon 2021 and achieved 9th place internationally.

Received Best Design Award (2022) for robotic system integration and execution quality.

Co-invented a patented robotic mechanism (Indian Patent: IN202321008858).

Co-developed a pellet-based 3D printing system recognized with the Vishvakarma Award at IIT Delhi.